

基于物理信息神经算子的Grad-Shafranov方程快速求解方法研究

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摘要: 快速求解托卡马克等离子体平衡对实时控制和平衡重建具有重要意义。描述轴对称环形等离子体平衡的Grad-Shafranov方程, 其传统Picard迭代有限差分数值解法虽精度高但求解耗时较长, 难以满足实时场景需求。本文探索了物理信息神经算子替代传统求解器的可行性。工作从传统有限差分数值求解器和朴素物理信息神经网络方法出发, 在此基础上重点研究了采用U型傅里叶神经算子架构的固定边界物理信息神经算子方法。模型以主半径、小半径、拉长比、三角形变、等离子体电流、极向比压及剖面形状指数共八个物理参数为输入, 输出二维极向磁通分布。训练采用多目标物理损失函数, 包含掩码均方误差、GS方程残差、边界带约束、磁轴约束和等离子体电流积分约束。针对物理约束在训练初期引发发散的问题, 采用课程学习策略分阶段逐步引入物理损失项, 实现了稳定的训练收敛。消融实验验证了物理约束和课程学习策略各自对模型精度和训练稳定性的贡献。经过训练的物理信息神经算子模型单次推理可在毫秒量级完成, 相比传统Picard迭代求解器实现了数量级的加速, 且磁通分布与数值解保持高度一致。结果表明, 物理信息神经算子方法在保证求解精度的前提下可实现GS方程的快速求解, 为托卡马克等离子体平衡的实时重建与参数空间扫描提供了可行的技术路径。

关键词: Grad-Shafranov方程; 物理信息神经算子; U型傅里叶神经算子; 课程学习; 托卡马克平衡

A Physics-Informed Neural Operator Approach for Fast Solution of the Grad-Shafranov Equation

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Abstract: Fast solution of the Grad-Shafranov equation is essential for real-time plasma control and equilibrium reconstruction in tokamaks. While classical numerical solvers based on Picard iteration achieve high accuracy, their per-solve latency limits real-time applicability. This paper explores the feasibility of replacing traditional solvers with a physics-informed neural operator. Starting from a traditional finite-difference solver and a vanilla physics-informed neural network, we focus on a U-Fourier Neural Operator architecture for the fixed-boundary Grad-Shafranov problem. The model

maps eight physical parameters—major radius, minor radius, elongation, triangularity, plasma current, poloidal beta, and two profile shape exponents—to the two-dimensional poloidal flux distribution. Training employs a multi-objective physics-informed loss function comprising masked mean squared error, Grad-Shafranov equation residual, boundary band constraint, magnetic axis constraint, and plasma current integral constraint. A curriculum learning strategy progressively introduces physics loss terms to prevent the divergence commonly observed when physics constraints are applied from initialization. Ablation experiments quantify the contributions of physics-informed constraints and curriculum learning to model accuracy and training stability. The trained neural operator achieves inference on the millisecond scale, representing orders-of-magnitude acceleration over classical Picard solvers while maintaining high fidelity to numerical solutions. These results demonstrate that physics-informed neural operators offer a viable pathway toward fast Grad-Shafranov equilibrium computation for real-time tokamak applications and rapid parameter-space exploration.

Key words: Grad-Shafranov equation; physics-informed neural operator; U-Fourier neural operator; curriculum learning; tokamak equilibrium